



PRO 1

MJ2505 HT25 Practical Optimization of Energy Networks

Group 20- 1 Week from winter + 1 Week from summer

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Pandapower Network

Pandapower : a tool to simulate power network for grid planning, load analysis and short circuit. It can build buses, lines, loads and can run power flow in the power system.

Toolkit used in the project:

Pandapower.networks: a ready to use grid

- `Creat_cigre_network_mv`: a CIGRE standard medium voltage distribution grid, including buses, lines loads and transformers.

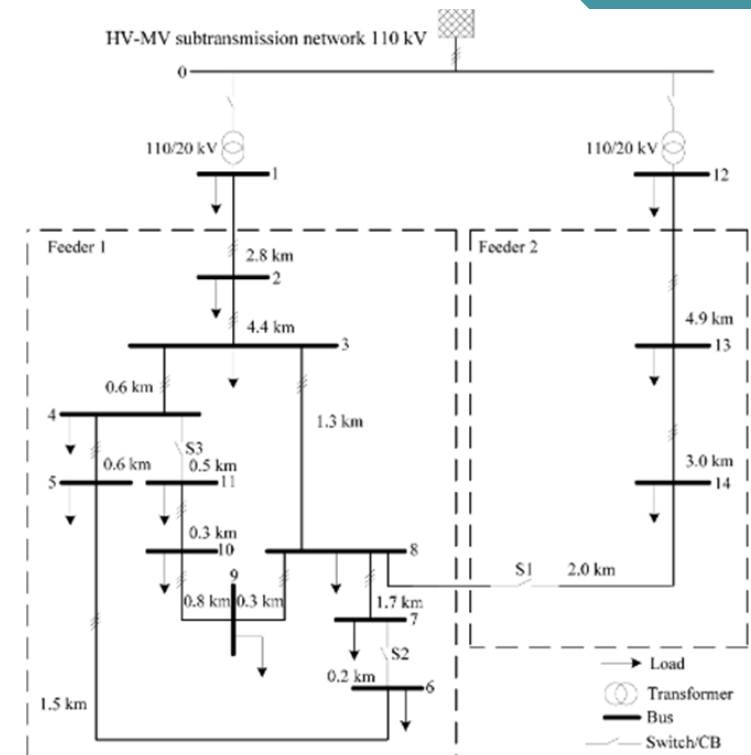
Pandapower.control :`ConstControl` to change loads hourly

Pandapower.timerseries: to run power flows over hours and days and log results through `OutputWriter` (voltages, line%, trafo%)



CIGRE Network Component

- Bus (15): Nodes for connections. Voltage (vn_kv), Type, Zone, Name
- Load (18): Power consumers. Active (p_mw), Reactive (q_mvar), Bus, Scaling
- Switch (8): Connectivity control. Bus/Element, Type (et), Status (closed)
- External Grid (1): Power source. Bus, Voltage (vm_pu), Angle, Short-circuit power
- Line (15): Power transmission. From/To Bus, Length, Type, Max Current
- Transformer (2): Voltage conversion HV/LV Bus, Power (sn_mva), Voltages, Losses, Tap settings



Step 1: Baseline network health Check

Method: Run PF on net_base with 50% Active and reactive power of loads and check

- Voltages (res_bus.vm_pu) max/min
- Line peak loadings (res_line.loading_percent, res_line.i_ka)
- Transformer loadings (res_trafo.loading_percent)

Criteria: Voltage: 0.90–1.10 pu (and stricter 0.95–1.05 pu cases), Thermal loading: lines $\leq$ 100%, transformers $\leq$ 100%.

Result:

- All the voltages are inside the range
- All loading percentages are below 45.13%, so no overloading risks for the lines
- All transformer loading percentages are below 50%

Step 1.2,1.3: Baseline N-1 Contingency Analysis

Objective: Check N-1 contingency on baseline network stability.

Steps:

1. Close all switches
2. Outage one line (not consider Line 6-7, Line 11-4, Line 14-8 -> switches)
3. Run PF, read results (res_bus, res_line, res_trafo).

Limits: Voltage band: 0.90–1.10 pu, Line & transformer loading: $\leq 100\%$

4. Loop over all lines and mark as critical lines if out of limits

```
cont_analysis(net_contanalysis1, v_min=0.9, v_max=1.1, line_loading_limit=100, trafo_loading_limit=100)
```

No contingencies found



Step 2.1: Develop Depot placement

- Extract individual peak value for each depot [MW]: ICA=1.50, Mathem=1.01, Postnord=1.42, Airmee=0.46

Sort Buses

Load Addition

Validation

find_suitable_buses:

- Sorts buses based on `vm_pu`
- Discards bus from the list if `line.from_bus` or `line.to_bus` have a loading % > 85%

add_single_load:

- Adds load to the first bus of the “find_suitable_buses” list
- Considered a power factor of 0.95

validate_network_after_changes:

- Runs PF
- Checks system ($\max(\text{line/trafo}) \leq 100\%$; $0.9 \leq \text{vm_pu} \leq 1.1$)
- Runs contingency analysis

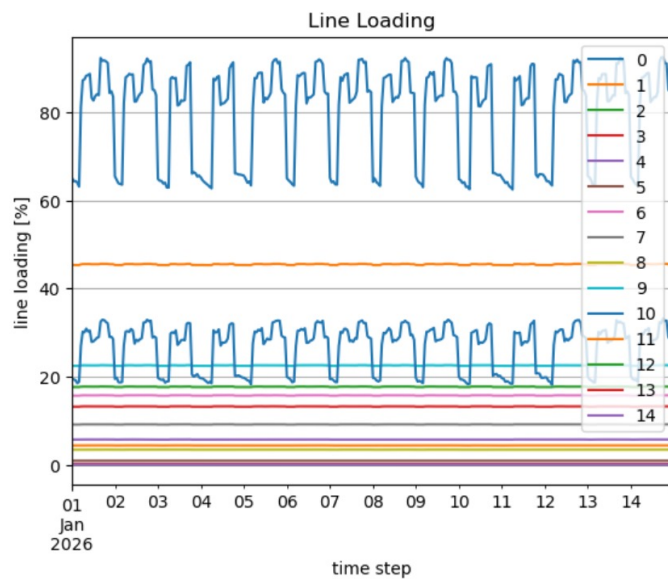
validate_network_no_contingency:

- Same but without contingency analysis

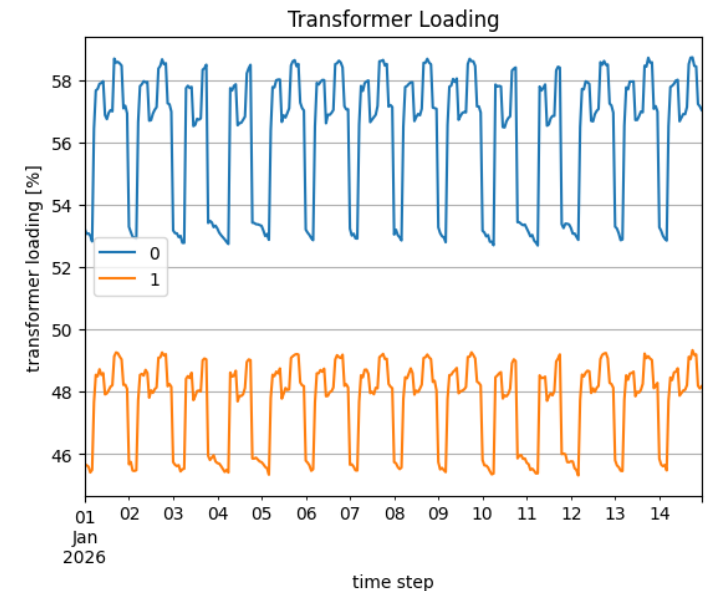
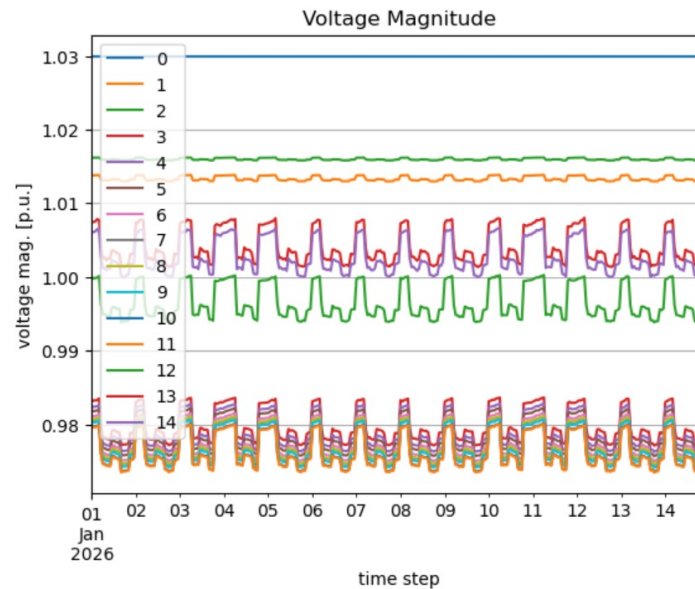
```
{'ICA': 13, 'Airmee': 13, 'Mathem': 2, 'Postnord': 2}
```

Step 2.2/2.3: Time-Series Simulation Setup/Performance Analysis

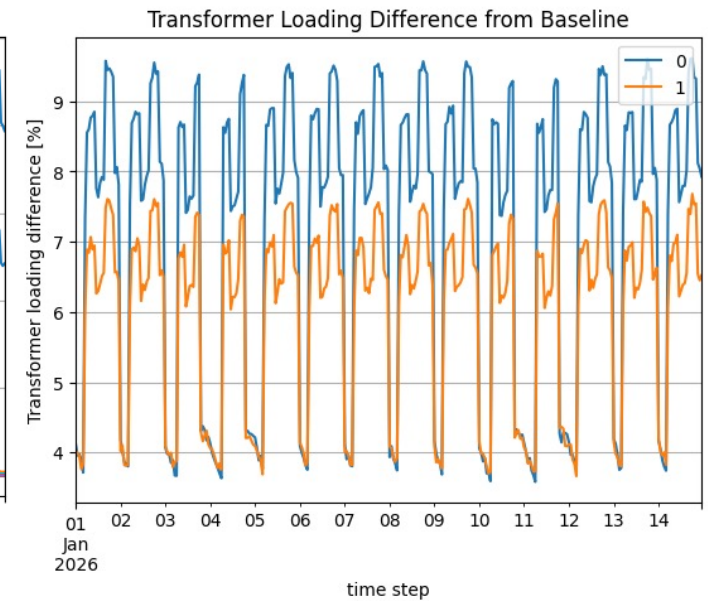
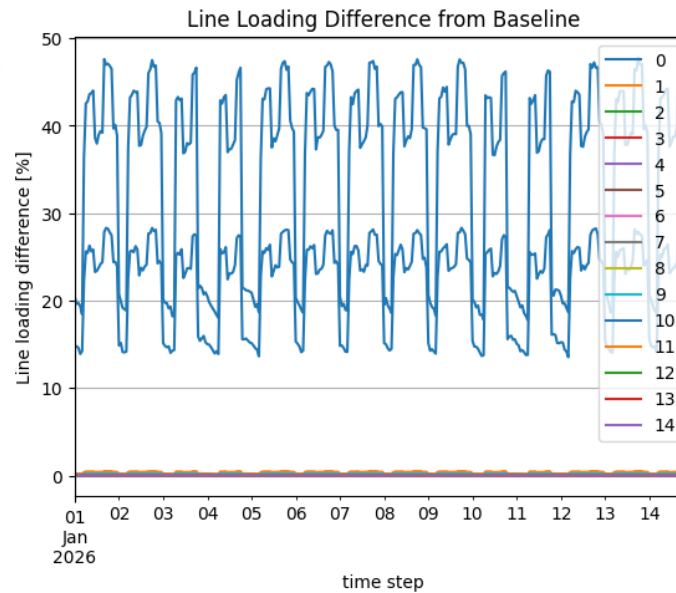
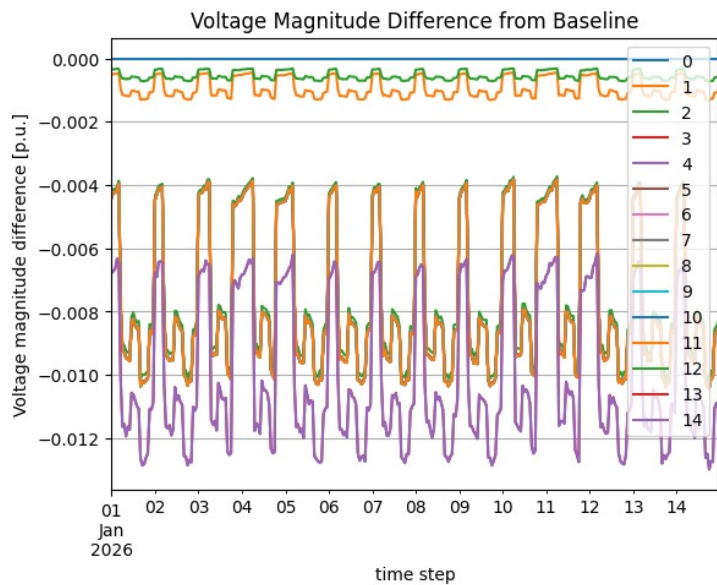
- Connect depot profiles to selected buses
- Run hourly load flow for 336 time steps (24 hours for 14 days)



Line 0: Line 1-2
Line 10: Line 12-13



Step 2.3: Comparison with Baseline Network



Step 2.4: Worst-hour contingency analysis

- Objective: select the most stressed hour and run N-1
- Pick worst hour by total depot load (MW) : 4.350126
- Worst hour: 2026-01-14 18:00 (+01:00)
- Add depot to the buses and run contingency analysis
- Limits: Voltage band: 0.90–1.10 pu, Line & transformer loading: $\leq 100\%$

```
Critical contingency found when removing 'Line 1-2':  
- Max loading on other lines: 'Line 12-13' at 116.4 %  
- Voltages: Bus 1 = 1.017 pu, Bus 2 = 0.876 pu  
- Max Transformer loading: 72.4 %  
Critical contingency found when removing 'Line 12-13':  
- Max loading on other lines: 'Line 1-2' at 152.2 %  
- Voltages: Bus 12 = 1.017 pu, Bus 13 = 0.902 pu  
- Max Transformer loading: 71.1 %  
Critical contingency found when removing 'Line 13-14':  
- Max loading on other lines: 'Line 1-2' at 103.0 %  
- Voltages: Bus 13 = 0.998 pu, Bus 14 = 0.960 pu  
- Max Transformer loading: 61.0 %
```

Step 2.4: Time-series contingency analysis

- Run the time-series contingency analysis (336 times per line shutted down)
- Limits: Voltage band: 0.90–1.10 pu, Line & transformer loading: $\leq 100\%$
- N-1 analysis for each line (excluding switches – 12, 13, 14)

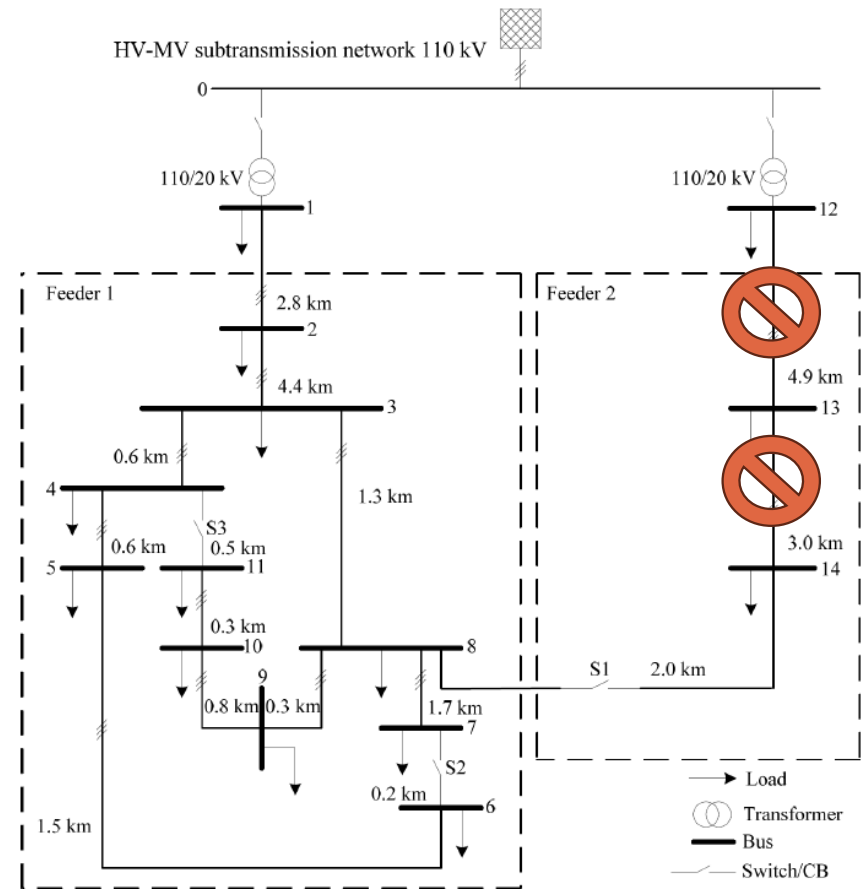
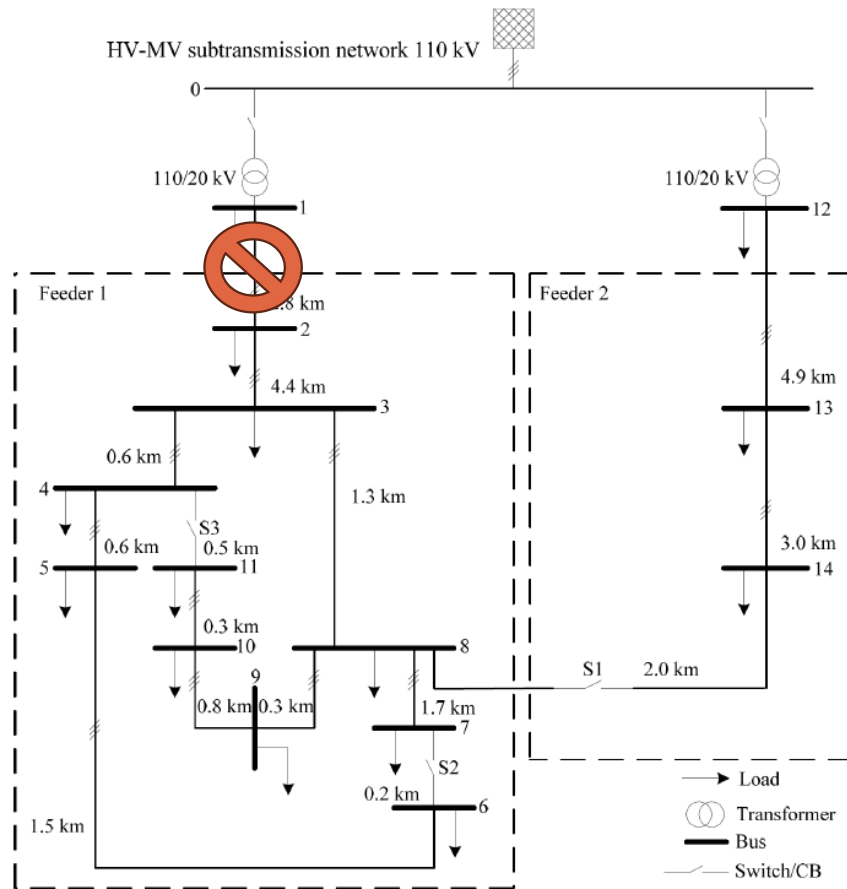
	outaged_line_name	max_other_line_name	count
0	Line 1-2	Line 12-13	238
1	Line 12-13	Line 1-2	336
2	Line 13-14	Line 1-2	62

Step 2.5: Time-series contingency analysis (stricter limits)

- Limits: Voltage band: 0.95–1.05 pu

	outaged_line_name	max_other_line_name	count
0	Line 1-2	Line 12-13	336
1	Line 12-13	Line 1-2	336
2	Line 13-14	Line 1-2	62

Step 2.5: Contingency Analysis considerations



Step 3: Fleet Configuration

With schedule generator configuration, the fleet table is created.

Company	Fleet size	Vehicle mix	Company type
ICA	10 EVs	Renault 5 / Toyota 5	Distribution
Mathem	7 EVs	Renault 4 / Toyota 3	Food
Postnord	8 EVs	Renault 4 / Toyota 4	Mail
Airmee	4 EVs	Renault 2 / Toyota 2	Mail



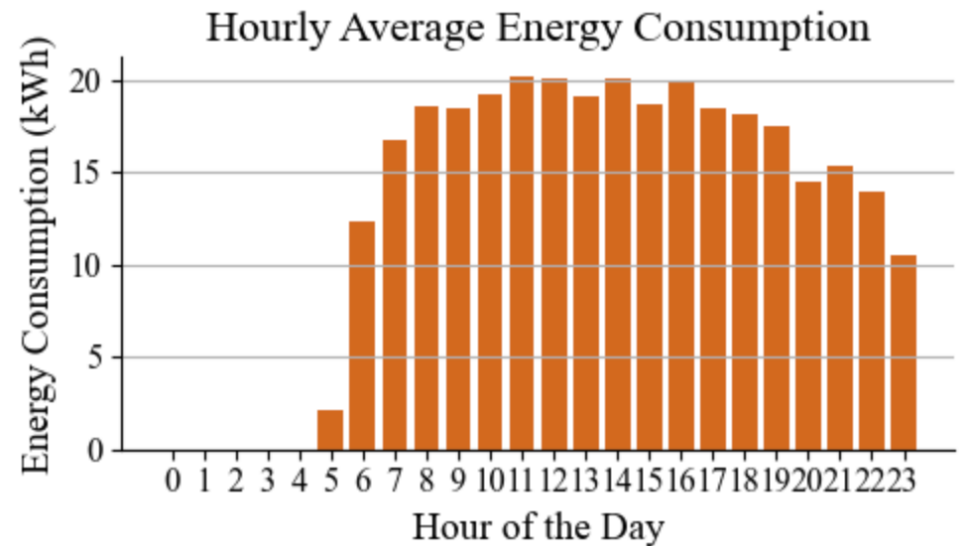
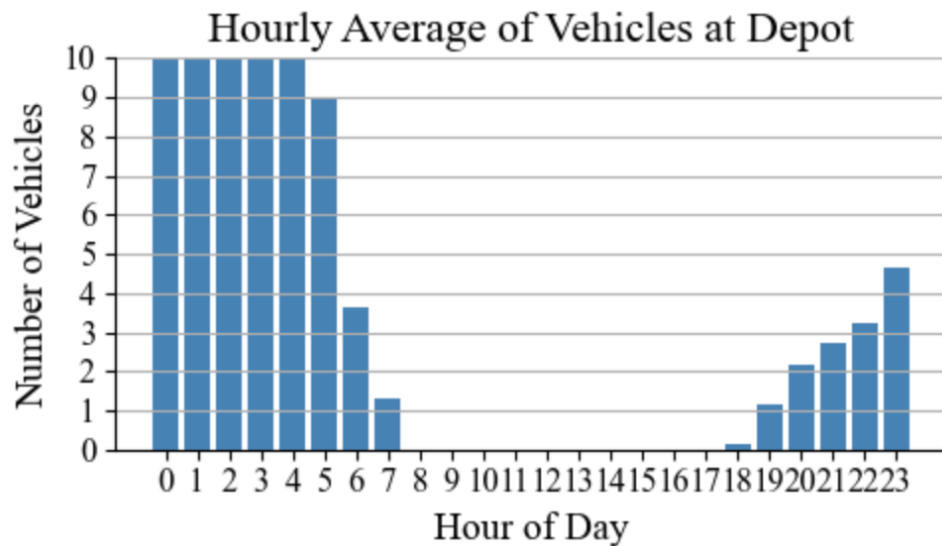
Done with efleetscheduler packages:

- Generator of synthetic duty schedules for an EV fleet



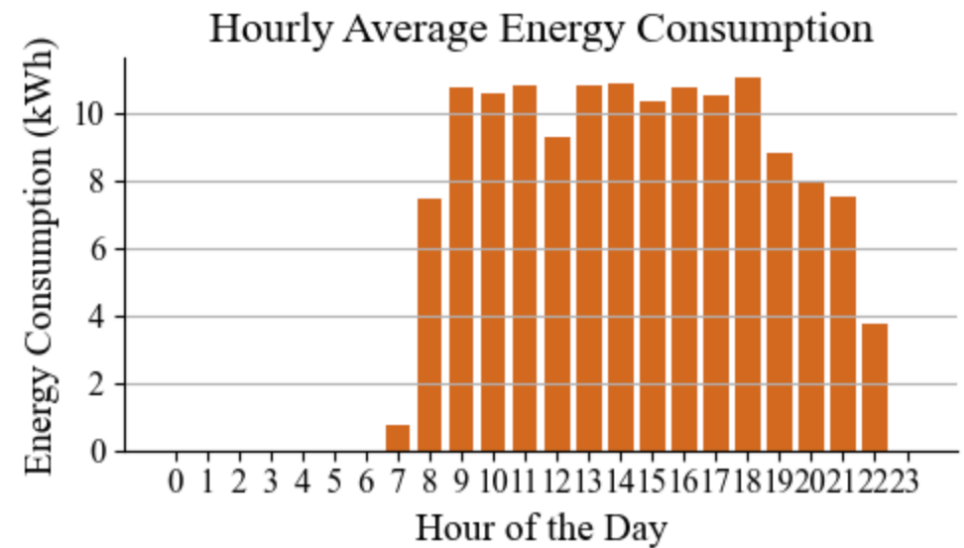
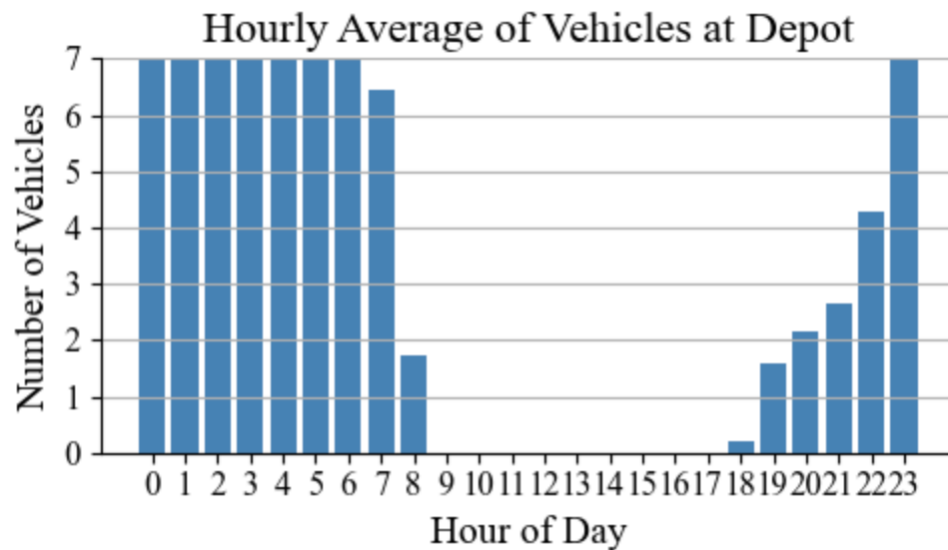
Step 3: Fleet Configuration

Company: ICA



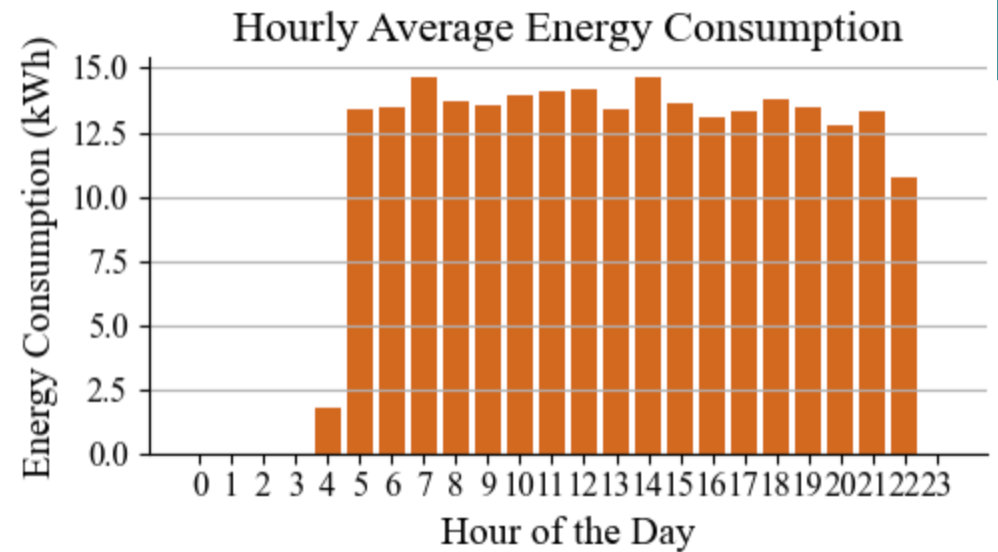
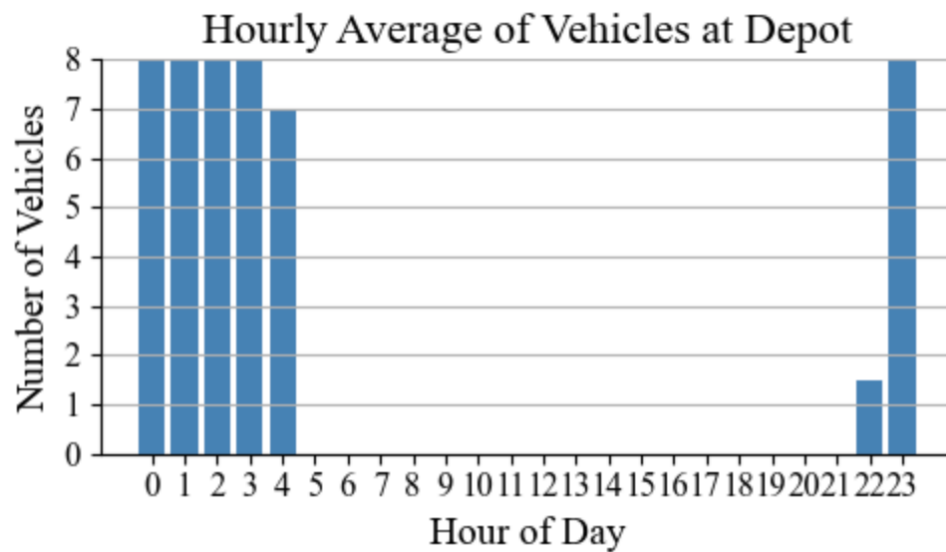
Step 3: Fleet Configuration

Company: Mathem



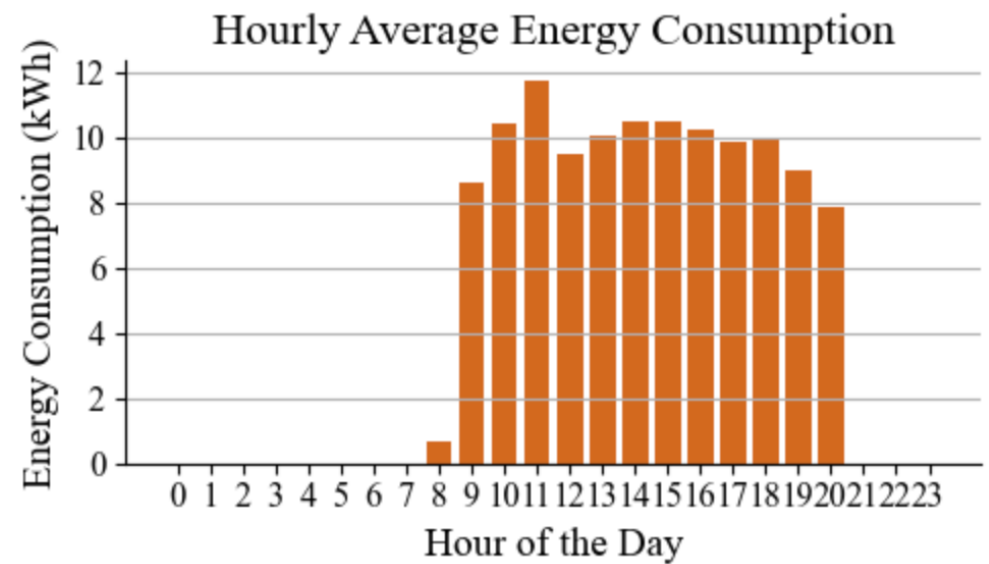
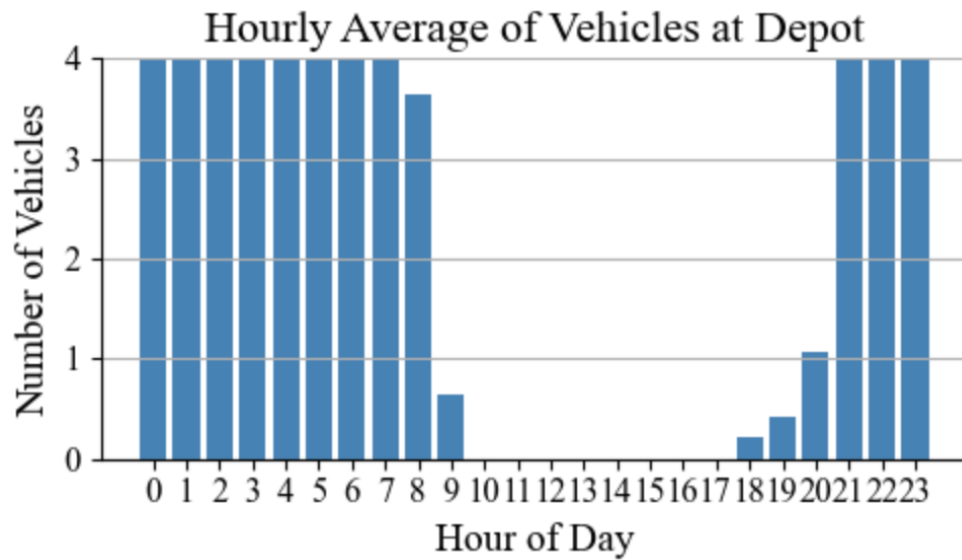
Step 3.1: Fleet Configuration

Company: PostNord



Step 3: Fleet Configuration

Company: Airmee



Step 3: Charging Design Decision



What charging infrastructure? Why?

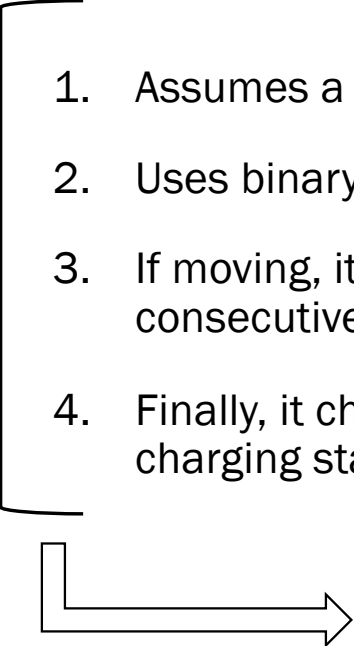
First approach assumptions:

- One charger per vehicle → More flexible for further optimization
- Same type of charger for all the depots → Vehicles with same characteristics
- Fast AC recharging point, triple phase – $P = 22 \text{ kW}$ → Be sure to match the demand and simplicity for further improvements

Step 3: Modelling dumb charging

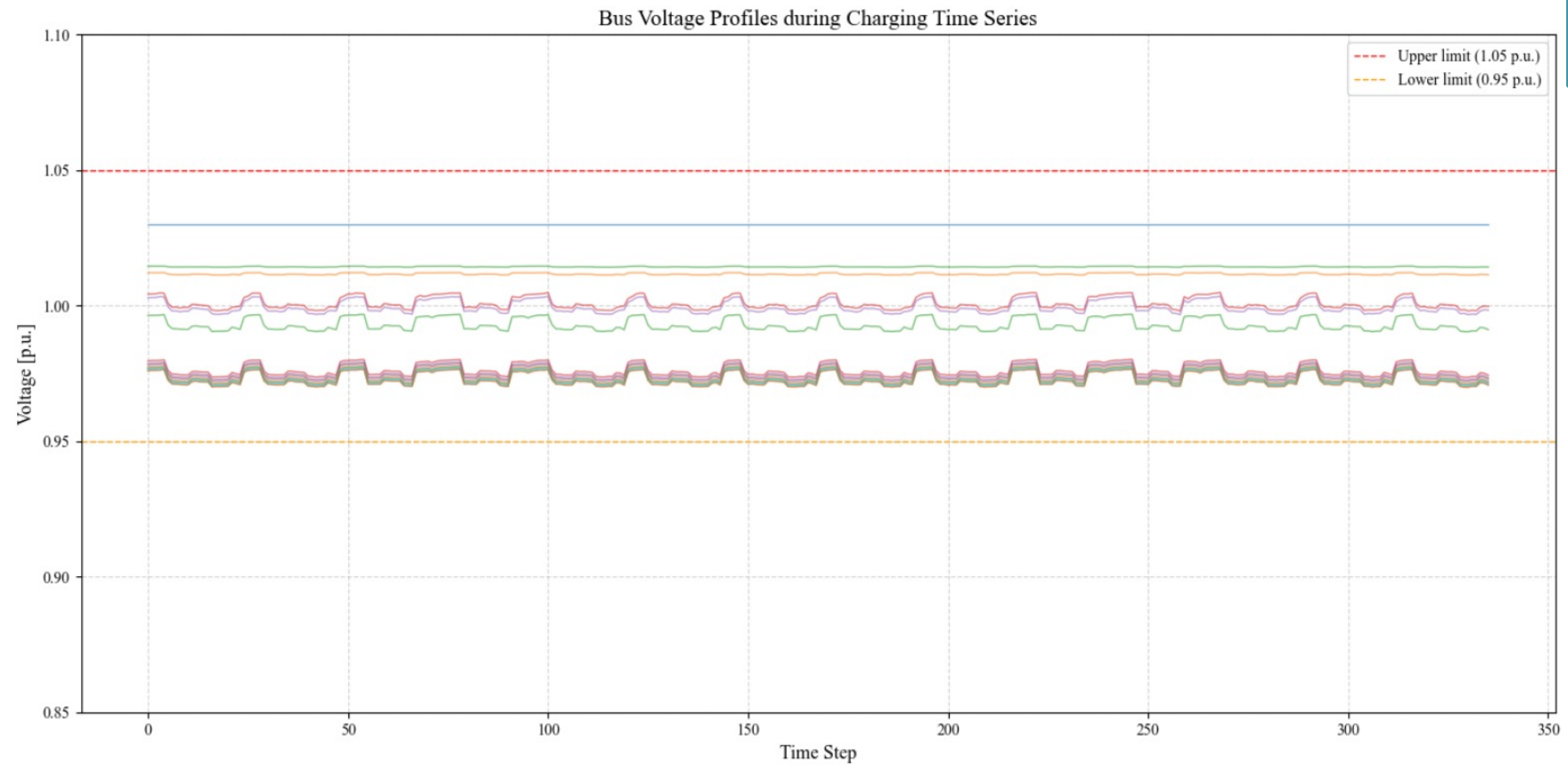
How is the model charging the EVs?

1. Assumes a SOC of 100% at the first hour of the dataset per vehicle
2. Uses binary values to understand if vehicle is moving (0) or not (1)
3. If moving, it consumes. The model sums the consumptions (kWh) for each vehicle - for each hour - consecutively moving
4. Finally, it charges the sum in full power (22 kW) or partially if not needed, from the first hour at the charging station

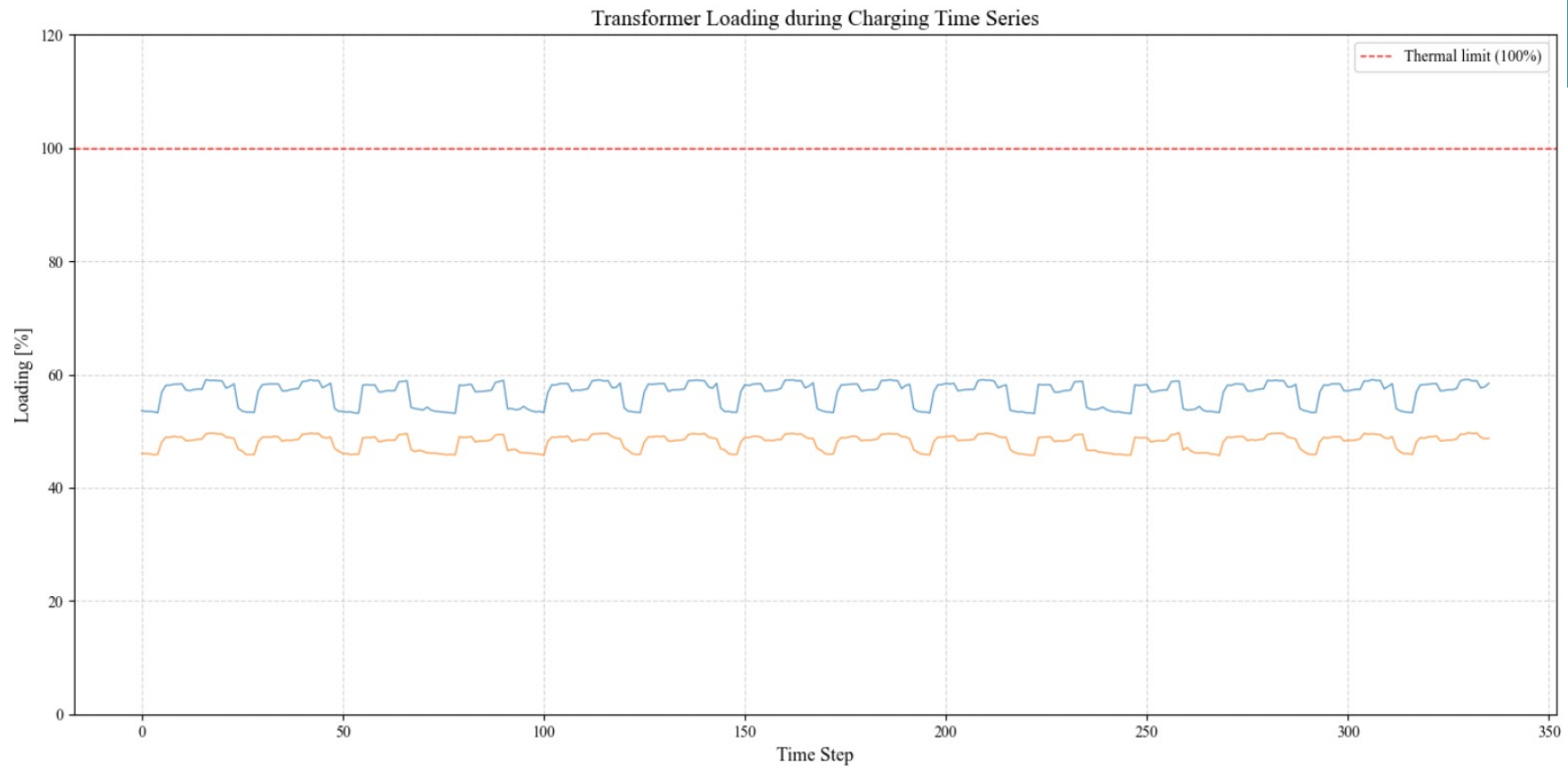


```
calculate_charging_power(company, env_config)
```

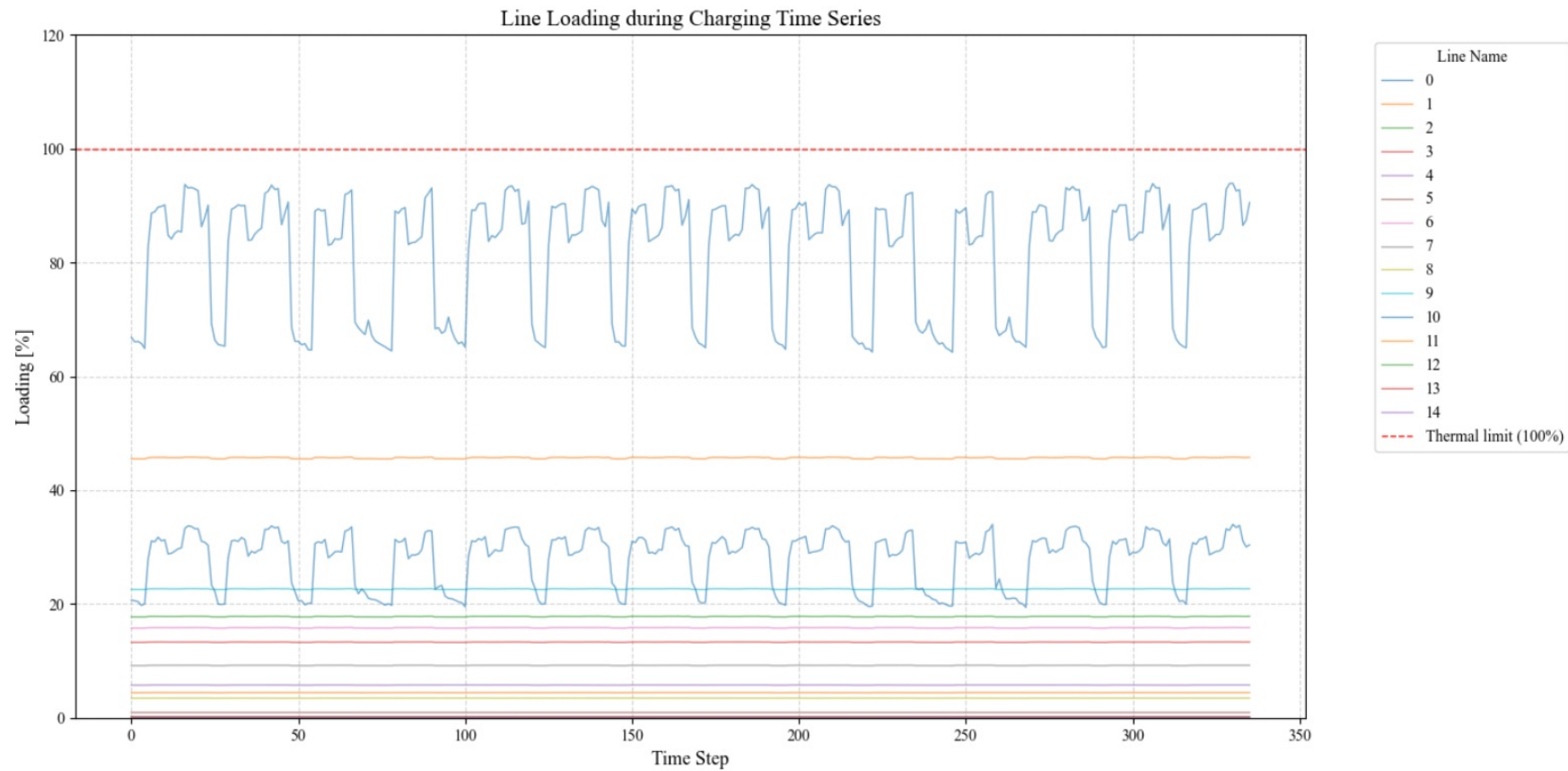
Step 3: Results dumb charging



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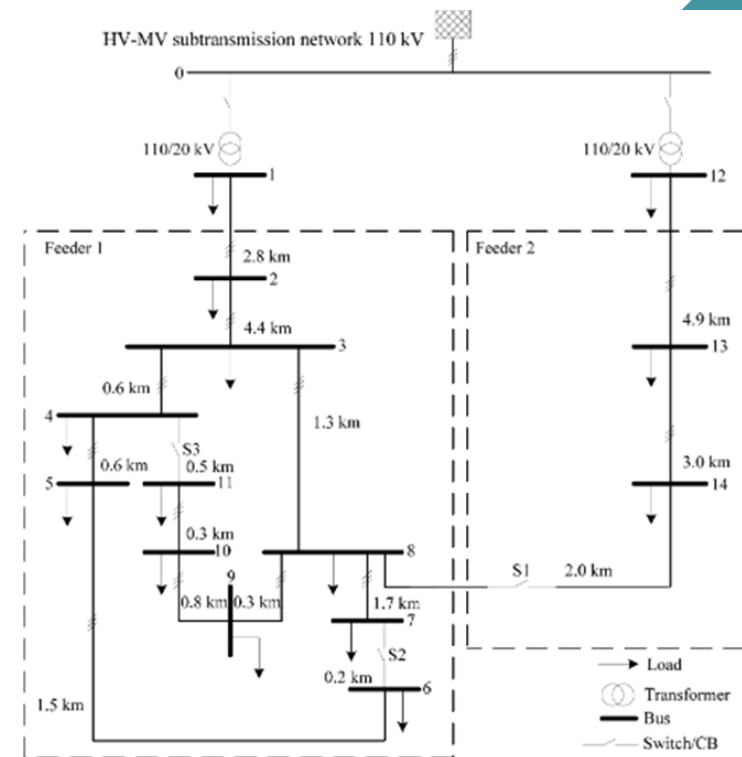
Step 3: Results dumb charging



Step 3: Hourly Contingency Analysis

	outaged_line_name	max_other_line_name	count
0	Line 1-2	Line 12-13	143
1	Line 12-13	Line 1-2	246

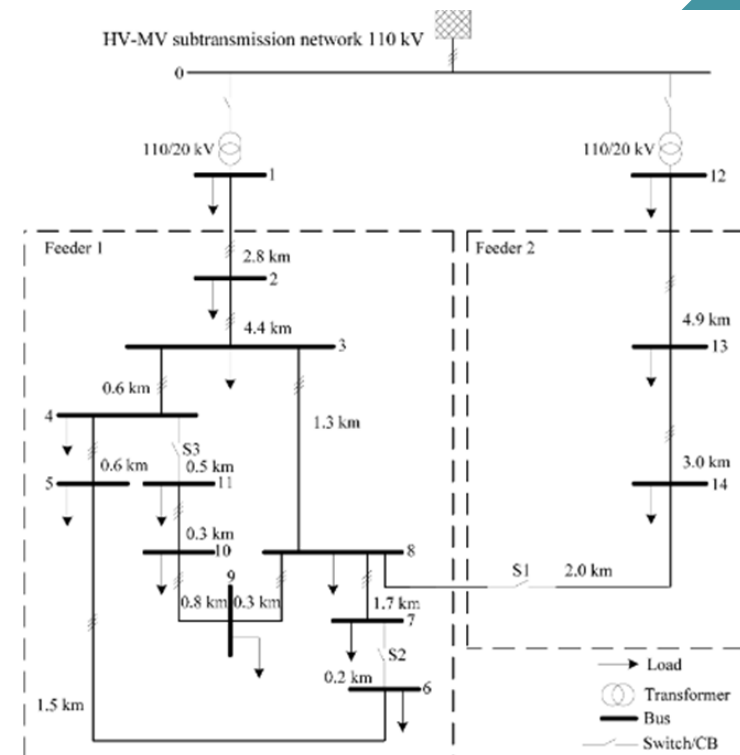
- maximum voltage 1.1 pu, minimum 0.9 pu, maximum line loading 100 %
- 336 are the total timesteps of the datasets
- When lines 1-2 and 12-13 are outaged, lines 12-13 and 1-2 respectively are in overloading



Step 3: Hourly Contingency Analysis

	outaged_line_name	max_other_line_name	count
0	Line 1-2	Line 12-13	336
1	Line 12-13	Line 1-2	247

- maximum voltage 1.05 pu, minimum 0.95 pu, maximum line loading 100 %
- 336 are the total timesteps of the datasets
- When lines 1-2 and 12-13 are outaged, lines 12-13 and 1-2 respectively are in overloading



Step 3: Charging Design Decision

What could be done better?

- Fast AC recharging point, triple phase – $P > 22 \text{ kW}$ —————> Slow chargers of roughly 4 kW
- Reducing number of chargers
- Optimize the charging strategy —————> Market and losses driven
- Ultimately reinforce line 1-2 and 12-13 if contingencies problems are not solved with optimization

Step4.1: Optimization Model with 1 Week from Winter & Summer Electricity Price

Inputs

Parameter	Description
I	Set of time intervals
Price	Vector of electricity prices [€/kWh] for each time interval
E_sess	Total required charging energy for the session [kWh]
Charger_Power	Maximum charging power of the charger [kW]



Charging Session

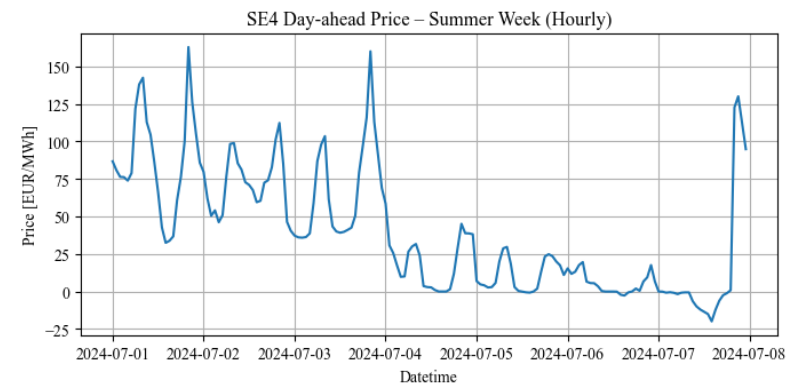
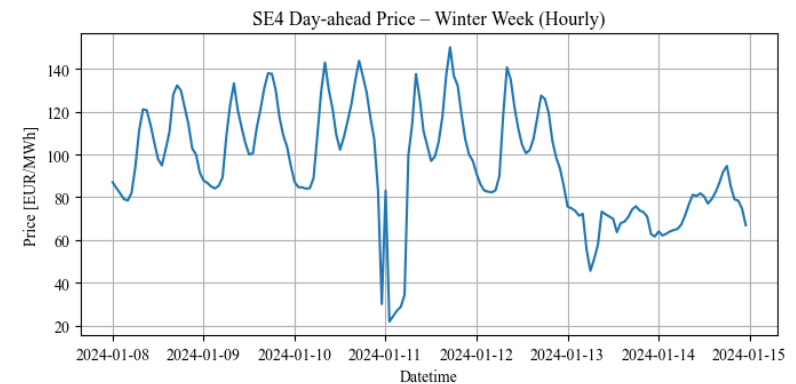
1	2	3	4	5	6	7
0	1	1	1	1	1	0

Step4.1: Optimization Model with 1 Week from Winter & Summer Electricity Price

Inputs

Parameter

Parameter	Description
I	Set of time intervals (e.g., hours of the day)
Price	Vector of electricity prices [€/kWh] for each time interval
E_{sess}	Total required charging energy for the session [kWh]
Charger_Power	Maximum charging power of the charger [kW]



Step4.1: Optimization Model with 1 Week from Winter & Summer Electricity Price

Inputs

Parameter	Description	
I	Set of time intervals (e.g., hours of the day)	
Price	Vector of electricity prices [€/kWh] for each time interval	
E_sess	Total required charging energy for the session [kWh]	How much energy did the vehicle consume while not charging <code>E_sess_local = float(consumption.iloc[prev_end:start].sum())</code>
Charger_Power	Maximum charging power of the charger [kW]	

Step4.1: Optimization Model with 1 Week from Winter & Summer Electricity Price

Inputs

Parameter	Description
I	Set of time intervals (e.g., hours of the day)
Price	Vector of electricity prices [€/kWh] for each time interval
E_sess	Total required charging energy for the session [kWh]
Charger_Power	Maximum charging power of the charger [kW]

→ **22 kW** design choice

Step4.1: Optimization Model with 1 Week from Winter & Summer Electricity Price

Decision variable: $P_{v,t}$ = charging power for vehicle v in time slot t [kW] (bound to be between 0 and max charging power (determined by the charging point chosen))

Constraint: total energy charged must equal required session energy ($E_{\text{sess}} / \text{CS_losses}$)

Objective : $\min \sum_i \text{Charge_hourly}_i \times \text{Price}_i$

Output: Optimal charging profile for the session and total charging cost

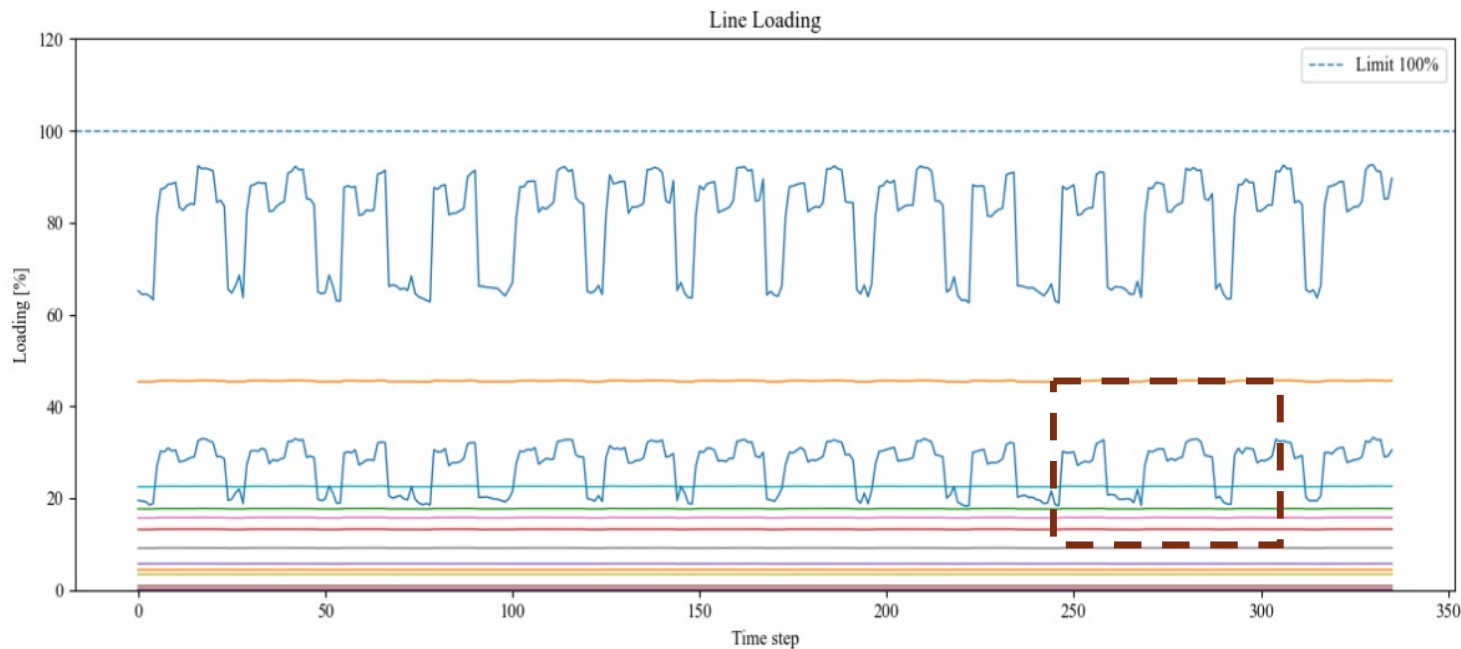
For each depot

For each EV

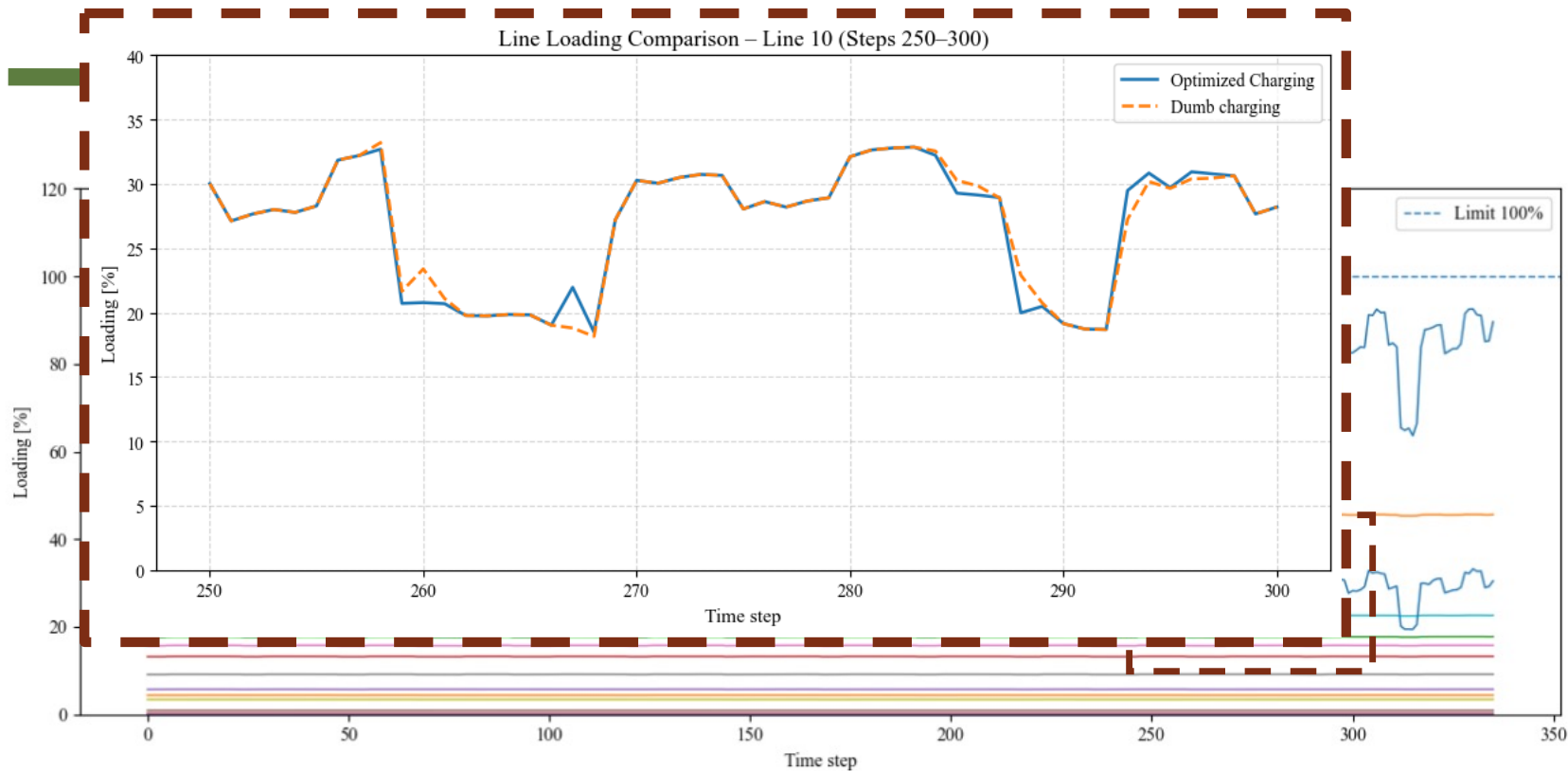
For each session

Optimize() session with linear optimization

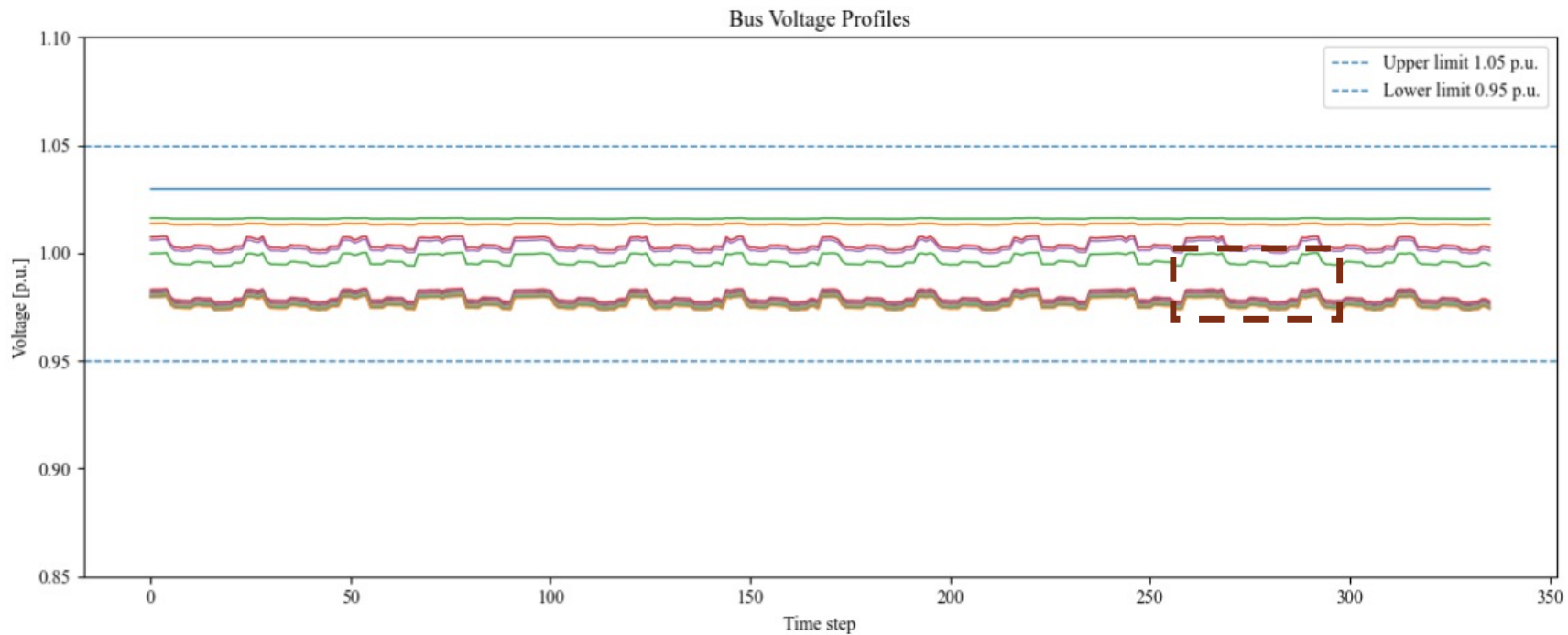
Step4: Price Optimized Model Network Results



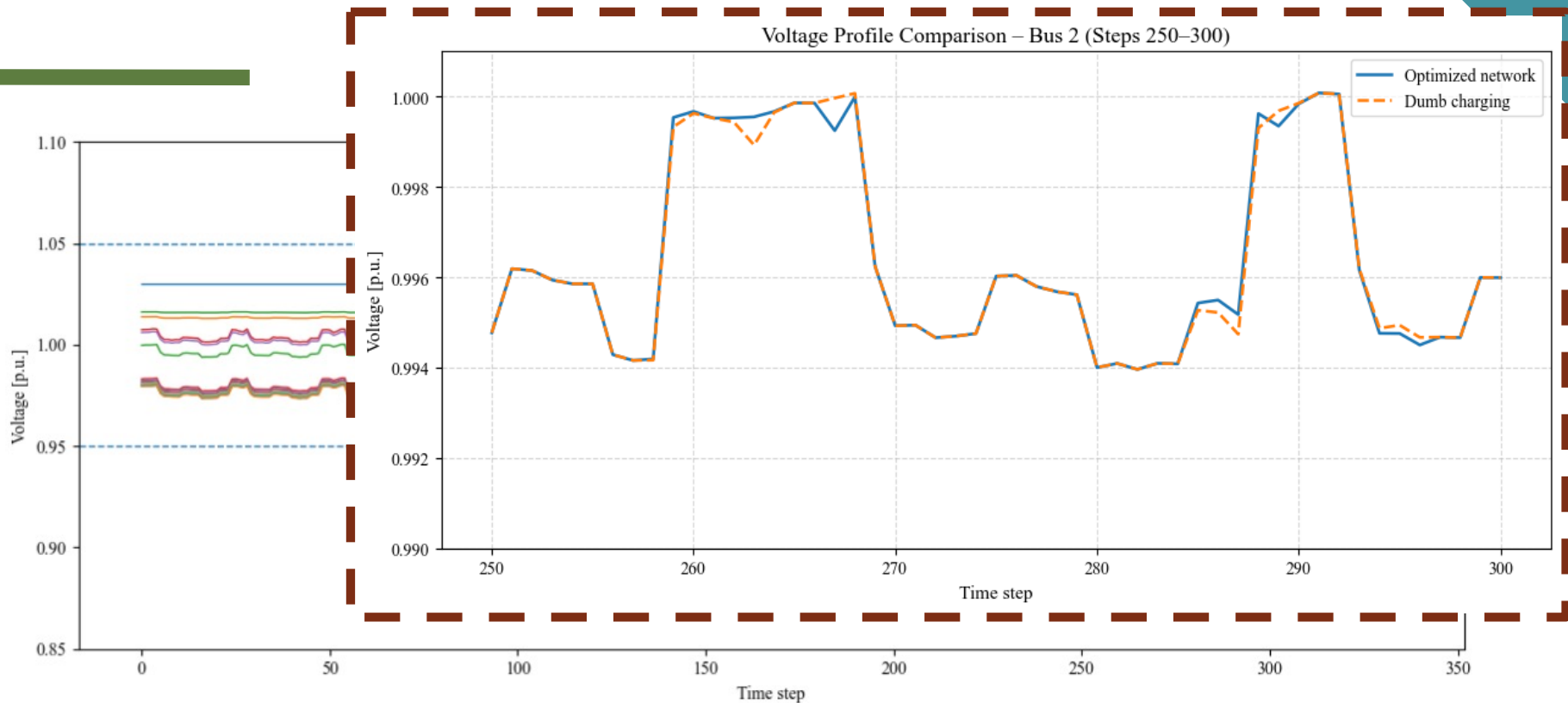
Step4: Price Optimized Model Network Results

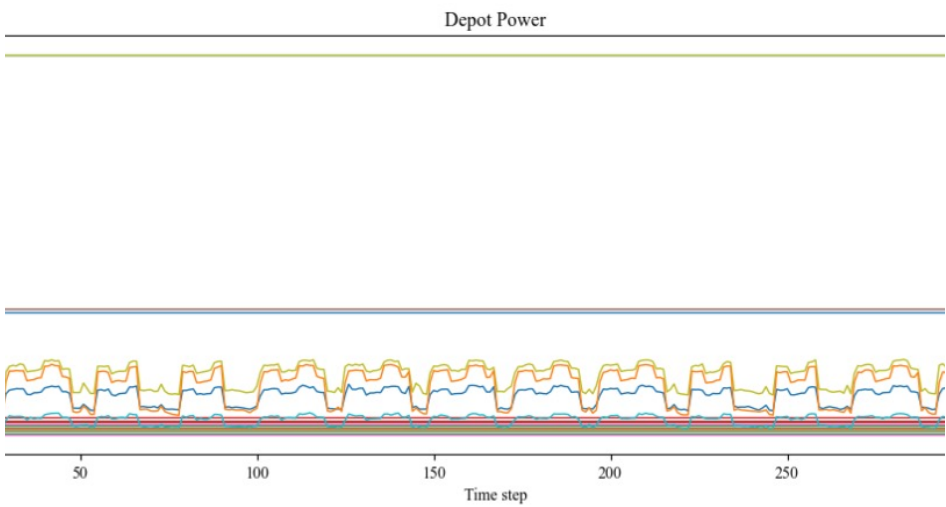
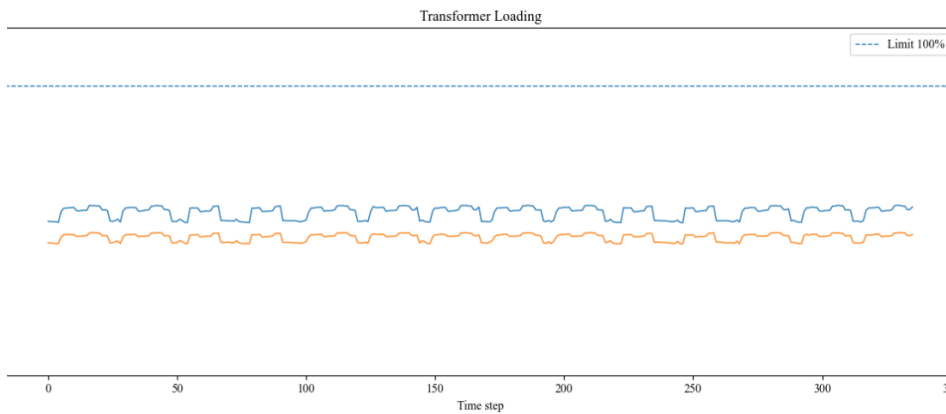


Step4: Price Optimized Model Network Results



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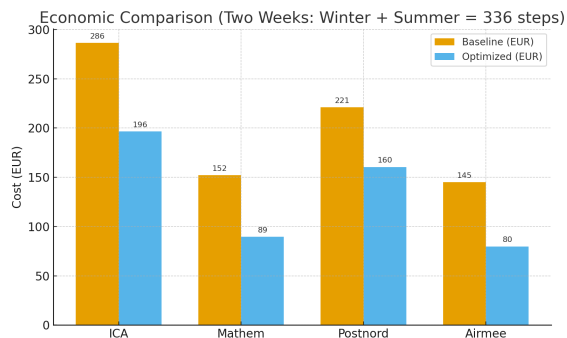




Step4: Price Optimized Model Network Results

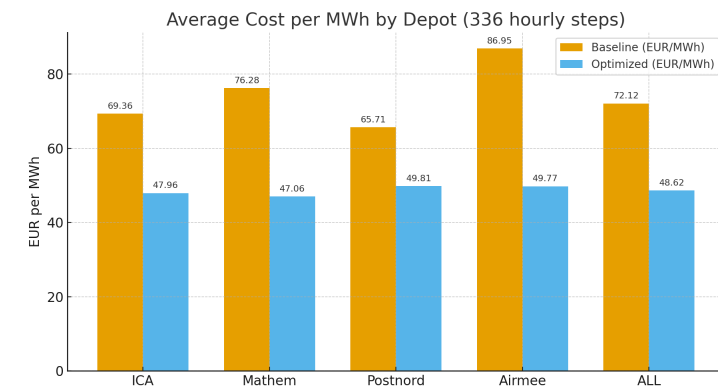


Step 4: Comparison, Dumb Charging vs Optimized Charging



Economic comparison (EUR)

Depot	Delta	Delta_%
ICA	-89.96	-31.41
Mathem	-62.63	-41.18
Postnord	-60.58	-27.41
Airmee	-65.35	-45.04

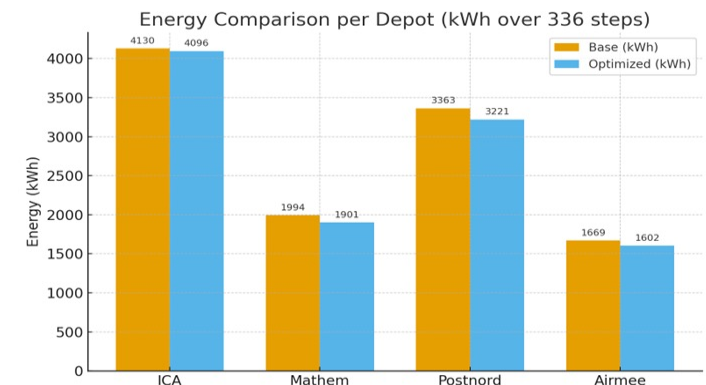


Average Cost by Depot (EUR/MWh)

Depot	Delta	Delta_%
ICA	-21.39	-30.85
Mathem	-29.21	-38.30
Postnord	-15.90	-24.20
Airmee	-37.18	-42.76
ALL	-23.50	-32.59

Energy comparison (kWh)

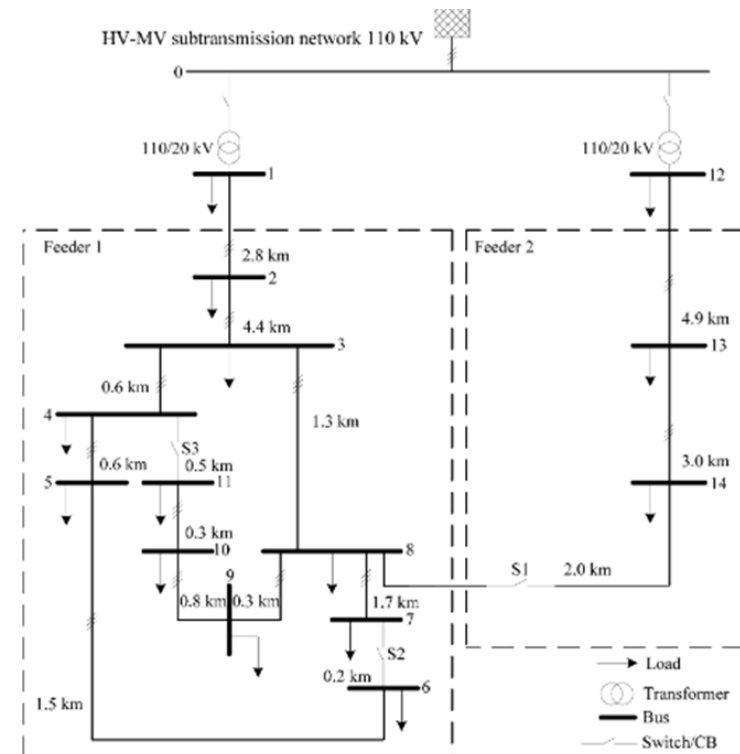
Depot	Delta
ICA	-33.53
Mathem	-92.95
Postnord	-142.64
Airmee	-66.39



Step 4: Losses – Contingency Analysis

```
outaged_line_name max_other_line_name count
0 Line 1-2 Line 12-13 131
1 Line 12-13 Line 1-2 243
```

- maximum voltage 1.1 pu, minimum 0.9 pu, maximum line loading 100 %
- 336 are the total timesteps of the datasets
- When lines 1-2 and 12-13 are outaged, lines 12-13 and 1-2 respectively are in overloading



Step 5: Losses – Driven Optimization

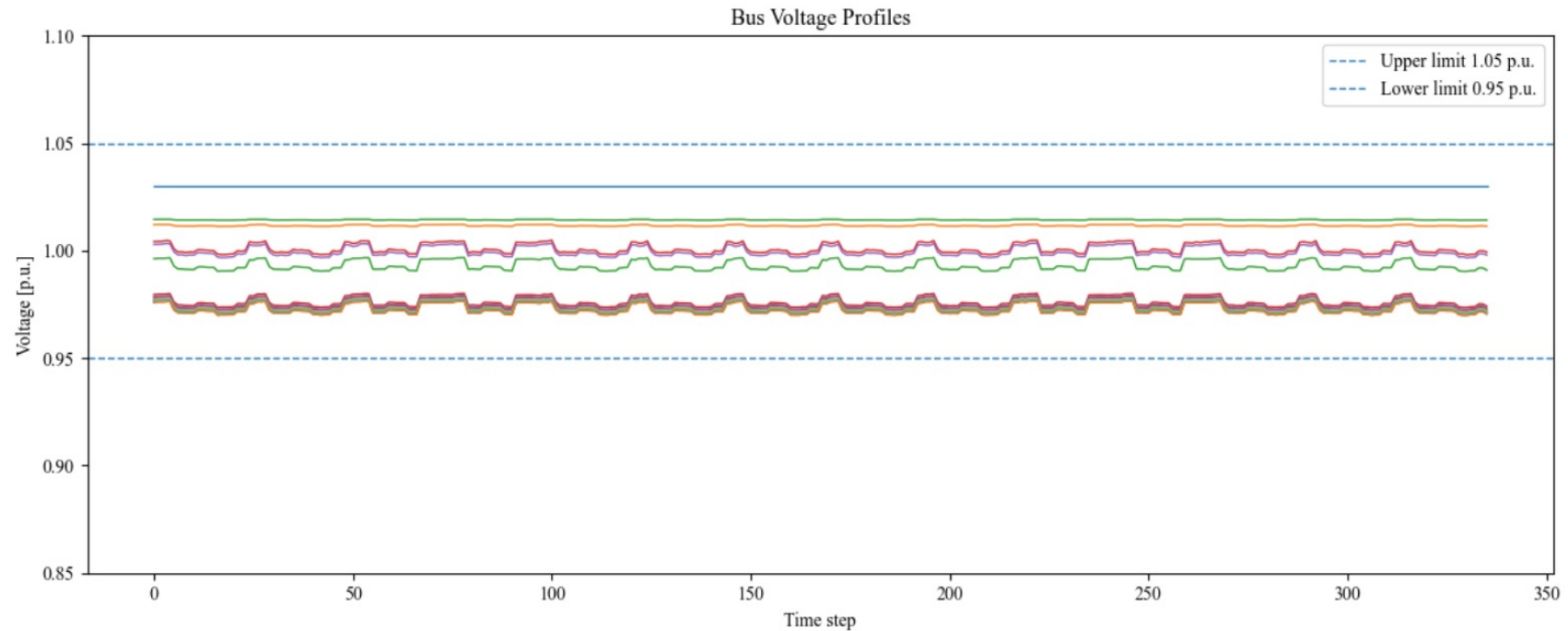
What difference from Market –Driven?

$$\text{minimize } \sum_{t=1}^{\text{hours}} \left(\sum_{i=0}^{i=FS} P_{i,t}^{EG} \right) \longrightarrow \text{Minimizing power injected and so losses}$$

$$\text{minimize } \sum_{t=1}^{\text{hours}} \left(\sum_{i=0}^{i=FS} P_{i,t}^{EG} \cdot \text{price} \right) \longrightarrow \text{Minimizing EUR/h}$$

Step 5: Losses – Driven Optimization

Load flow analysis and comparison to STEP 3



Step 5: Losses – Driven Optimization

Load flow analysis and comparison to STEP 3



Step 5: Losses – Driven Optimization

Load flow analysis and comparison to STEP 3

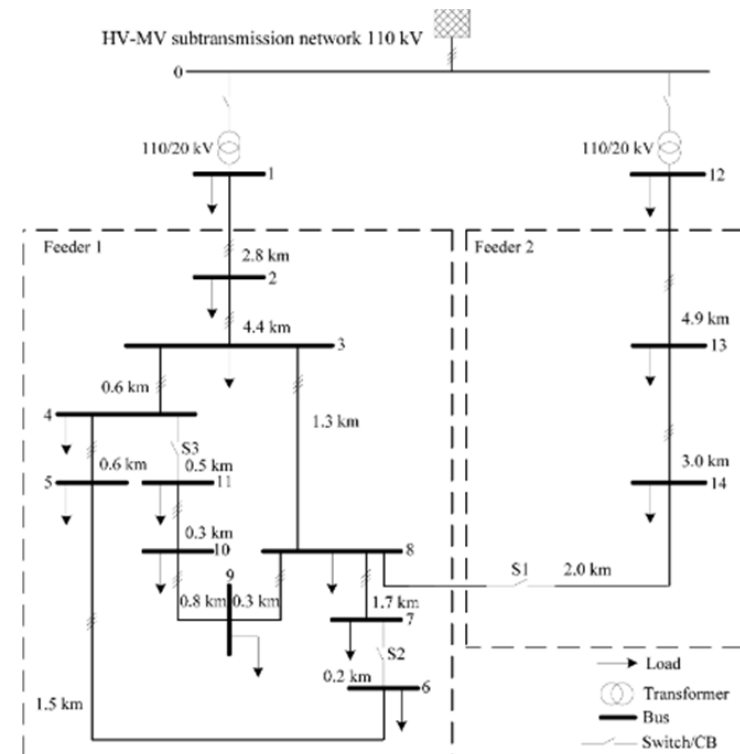


Step 5: Losses – Driven Optimization

Contingency analysis

	outaged_line_name	max_other_line_name	count
0	Line 1-2	Line 12-13	155
1	Line 12-13	Line 1-2	269
2	Line 13-14	Line 1-2	6

- maximum voltage 1.1 pu, minimum 0.9 pu, maximum line loading 100 %
- 336 are the total timesteps of the datasets
- When lines 1-2 and 12-13 are outaged, lines 12-13 and 1-2 respectively are in overloading



Step 5: Losses – Driven Optimization

Load flow analysis and comparison to STEP 3

```
=== NETWORK KPIs (first 336 steps) ===
```

	Vmin_pu	V<0.95_cnt	V>1.05_cnt	Line_max_%	Line>100_cnt	\
Baseline	0.96991	0	0	93.91115	0	
Optimized_v3	0.96991	0	0	93.91115	0	

	Trafo_max_%	Trafo>100_cnt	Energy_MWh	Delta
Baseline	59.11312	0	8642.30336	NaN
Optimized_v3	59.11312	0	8640.14266	NaN

Step 5: Losses – Driven Optimization

Load flow analysis and comparison to STEP 3

=== AVERAGE EUR/MWh (using 336 hourly steps) ===

	Baseline_EUR_per_MWh	Optimized_EUR_per_MWh	Delta_EUR_per_MWh	\
ICA	69.36	57.64	-11.72	
Mathem	76.28	58.95	-17.33	
Postnord	65.71	58.24	-7.46	
Airmee	86.95	80.30	-6.65	
ALL	72.12	60.71	-11.42	

Delta_%

ICA	-16.89
Mathem	-22.72
Postnord	-11.36
Airmee	-7.65
ALL	-15.83

Conclusion

Which is the best strategy? The baseline, price optimized or losses optimized charging?

N-1 contingency analysis of critical lines limits (0.9–1.1 pu; 100% trafos and lines loading)

Outage → Bottleneck	Baseline	Price optimization	Losses optimization
1-2 → 12-13	143	131	155
12-13 → 1-2	246	243	269
13-14 → 1-2	–	–	6
Total (count)	389	374	430
Δ vs Baseline	–	–15 (–3.9%)	+41 (+10.5%)

Conclusion



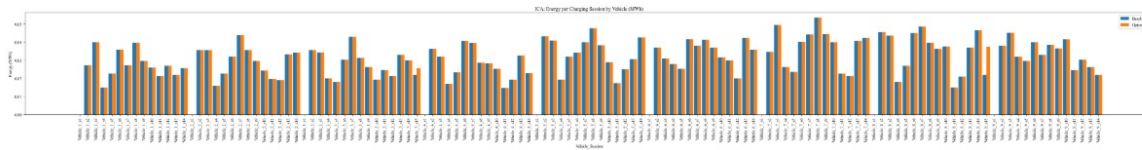
- Price optimization is retained to be the best strategy
 - Lowest price per MWh (€/MWh)
 - Lowest number of contingencies

Depot	Price €/MWh (Baseline)	Price €/MWh (Market - Driven)	Price €/MWh (Losses – Driven)
All	72.12	48.62	60.71

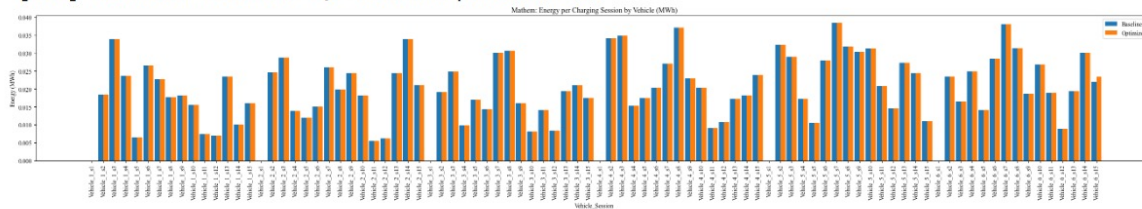
Thank you



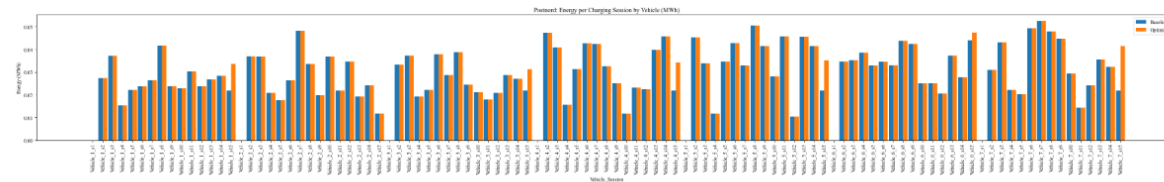
Step 4: Energy Comparison, Dumb Charging vs Optimized Charging



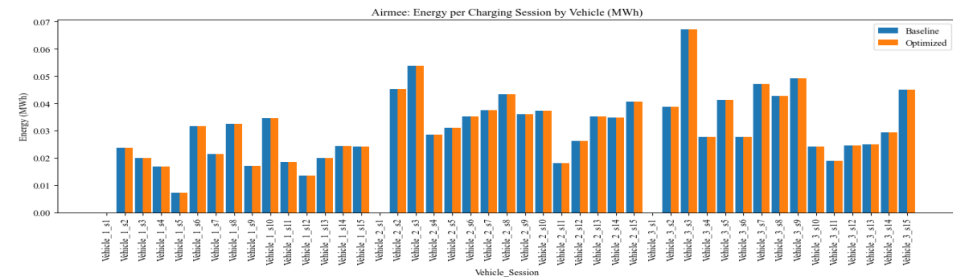
[ICA] Vehicles matched: 9, Sessions plotted: 128



[Mathem] Vehicles matched: 6, Sessions plotted: 90



[Postnord] Vehicles matched: 7, Sessions plotted: 105



[Airmee] Vehicles matched: 3, Sessions plotted: 45